Exercise 2 notes

Gravity as fxn of height from surface:

$$\vec{F}_g = \frac{GM_\oplus m}{r^2}\hat{r}$$

$$\vec{a}_g = \frac{GM_\oplus}{r^2}\hat{r} = \frac{GM_\oplus}{(R_\oplus + h)^2}\hat{r} = \frac{GM_\oplus}{(R_\oplus+h)^2}\frac{r_x \hat{i}}{|\vec{r}_x|} + \frac{GM_\oplus}{(R_\oplus+h)^2}\frac{r_y \hat{j}}{|\vec{r}_y|}$$

$$a_g = \frac{GM_\oplus}{(R_\oplus + h)^2}$$

if $h = 0$:

$$a_g = \frac{GM_\oplus}{R_\oplus^2} = 9.81 = g_0$$

where

$G = 6.6743 * 10^{-11}\ [Nm^2/kg^2]$

$M_\oplus = 5.9722 * 10^{24}\ [kg]$

$R_\oplus = 6.371 * 10^6\ [m]$

if $h << R_\oplus$:

$$a_g = \frac{GM_\oplus}{R_\oplus^2 + 2R_\oplus h + h^2}$$

$$= \frac{GM_\oplus}{R_\oplus(R_\oplus + 2h) + h^2} = \frac{GM_\oplus R_\oplus}{R_\oplus^2(R_\oplus + 2h) + h^2}$$

$$\lesssim \frac{g_0 R_\oplus}{R_\oplus + 2h} = \frac{g_0}{1 + 2h/R_\oplus}$$

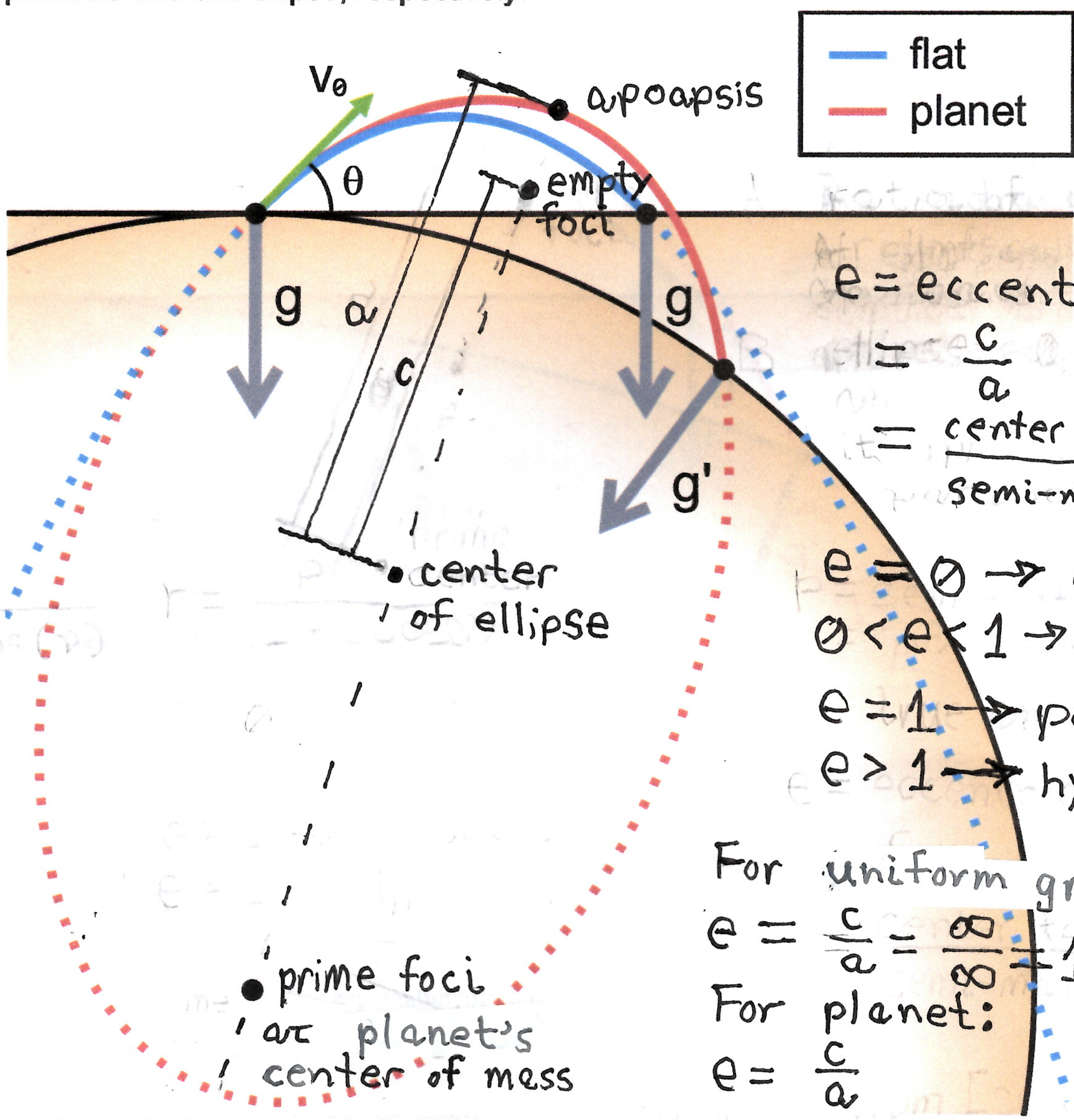
$$= \frac{g_0}{1+2h/R_\oplus} * \frac{1-2h/R_\oplus}{1-2h/R_\oplus} = \frac{g_0(1-2h/R_\oplus)}{1-4h^2/R_\oplus^2} \approx g_0(1-2h/R_\oplus)$$

which is a linearly decreasing fxn

Exercise 2 Notes

Visualization of Projectile Motion on a Planetary Scale

Plot of two ballistic trajectories with identical initial velocity, one with a uniform gravity field (blue) and one in the central potential field of a planet (orange). The trajectories are one parabola and one ellipse, respectively.



e = eccentricity

$= \dfrac{c}{a}$

$= \dfrac{\text{center to focus dist.}}{\text{semi-major axis}}$

$e = 0 \rightarrow$ circle

$0 < e < 1 \rightarrow$ ellipse

$e = 1 \rightarrow$ parabola

$e > 1 \rightarrow$ hyperbola

For uniform gravity:

$e = \dfrac{c}{a} = \dfrac{\infty}{\infty} = 1 \rightarrow$ parabola

For planet:

$e = \dfrac{c}{a}$

- Assumptions which yield parabolic trajectory:
  - Earth's gravitational field points straight down relative to launch location (i.e. prime foci at infinity)
- Assumptions which yield elliptical trajectory:
  - Acceleration vector points toward center of Earth (i.e prime foci NOT at infinity)
- Assumptions which affect impact point:
  - Shape of surface (flat, spherical, or oblate spheroid)